In this project, the product is a two-wheeled robot that can either operate autonomously or be remotely controlled. The robot is configured and monitored through a Human-Machine Interface (HMI) that we designed and developed ourselves.

The main objective of this project is to enable a user to explore an unknown environment using the robot and to create a map of this environment through our HMI. The robot is equipped with several sensors, including a LiDAR and a camera, which allow it to collect information about its surroundings. This data is then transmitted to the HMI, where it can be processed and displayed to the user.

The HMI provides the user with an overview of the robot's environment and allows them to monitor important information, control the robot remotely, and visualize the generated map. The project therefore combines robotics, autonomous navigation, data processing, image processing, and human-machine interaction.

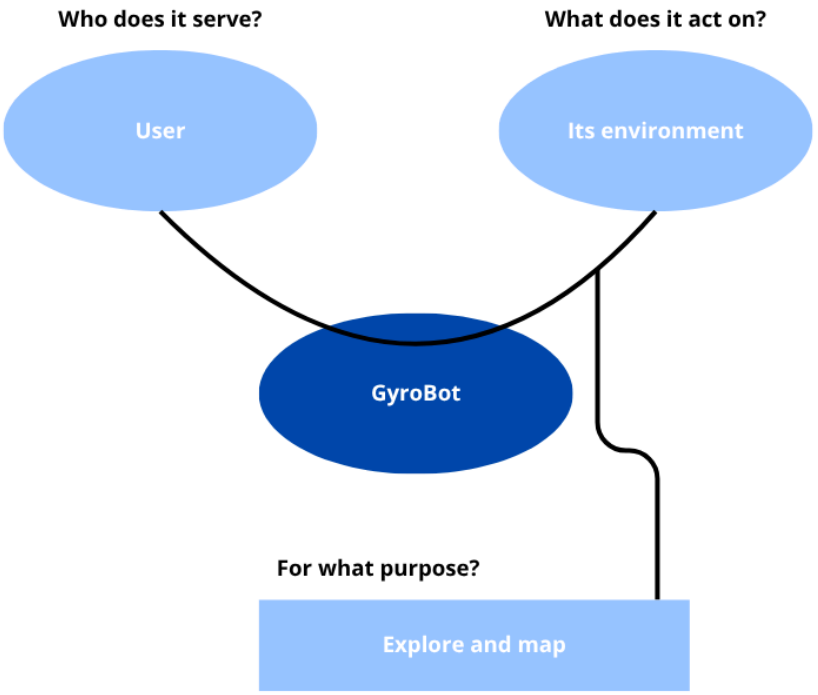


fig 1 : Functional Analysis Diagram (Horned Beast Diagram)

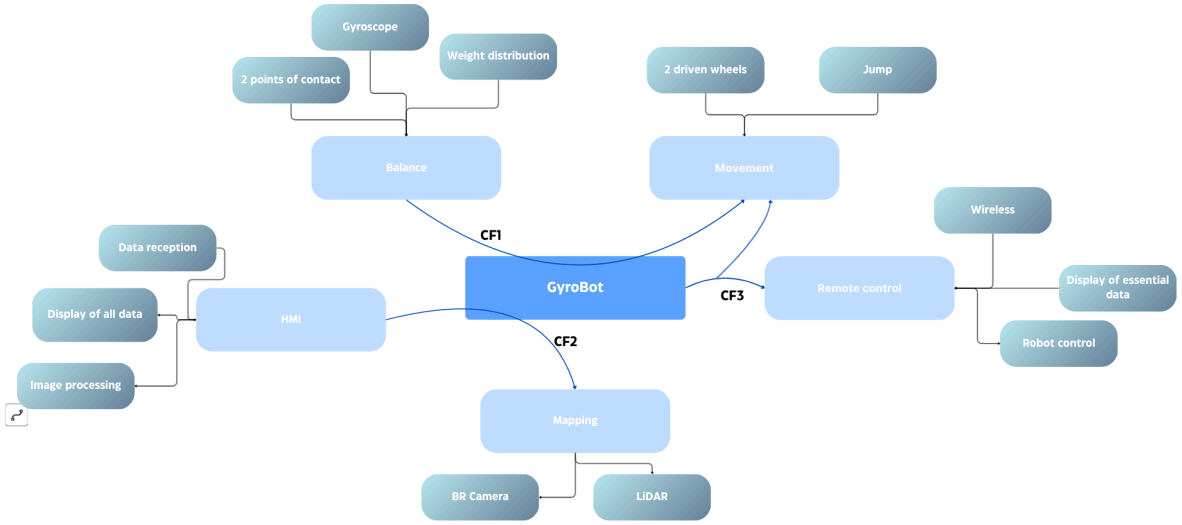


fig 2 : Functional Interaction Diagram (Octopus Diagram)

Reference	Constraint function	Level
CF1	Maintain balance during movement	High
CF2	Map its environment	Medium
CF3	Be remotely controlled	High